



Concrete vs. wood in buildings – An energy system approach

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ABSTRACT

Substitution between energy and CO₂ intensive materials is a potentially important climate mitigation strategy. We compare buildings with concrete frames and wooden frames concerning their life-time carbon dioxide emissions as well as their total material, energy and carbon dioxide costs. By using consistent energy systems scenarios meeting stringent targets for atmospheric CO₂ concentrations we investigate the impact of higher energy and carbon dioxide prices as well as of the availability of carbon capture and storage (CCS) technologies. We find that wooden frames cause lower carbon dioxide emissions given the prevailing energy system, but concrete frames obtain about the same emissions as the wood frame in a system where CCS is not used for wood incineration in the demolishing phase. The net present costs for the different buildings are also affected by the future energy supply system, even though the impact is small, especially compared to the total construction cost. We conclude that it is unclear whether wood framed buildings will be a cost-effective carbon mitigation option and that further analyses of costs should be performed before prescriptive materials policies are enforced in the buildings sector.

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1. Introduction

The discussion on climate change mitigation is often centred around potential changes of the energy system, such as substitution of fuels, enhancing energy efficiency, and developing carbon capture and storage technologies. However, reductions in greenhouse gas (GHG) emissions can also be achieved by changes in societies' production and use of materials. Energy intensive materials used in large volumes such as building materials are reasonably the most interesting ones in this respect. In Western Europe, the production of building materials accounts for as much as 8–12 percent of the total CO₂ emissions [1]. Even though energy use and CO₂ emissions from the building sector are dominated by the use phase, the production phase is not at all negligible, especially for low-energy buildings [2–4]. This study focuses on the cost-efficiency and carbon dioxide emissions from wood and concrete buildings in stringent CO₂ scenarios.

Disregarding the effects of overall volume decreases of materials turnover, for instance, as an effect of lower economic activity, the adaptation of the materials system towards lower GHG emissions may be done in at least two principally different ways. The first option is to decrease the GHG emissions from the production of the

specific materials. Analyses of these possibilities have been part of efficiency enhancement studies of the energy use in the industrial sector, a sector where energy use is dominated by the subsectors of primary conversion of materials, such as production of iron and steel, cement, chemicals, and pulp and paper (see e.g. de Beer et al. [5], Rootzén et al. [6], Neelis et al. [7]). The production processes themselves are here direct emitters, but the overall emissions are also strongly dependent on the GHG intensity of the energy supplied to the processes, such as electricity. The second option is by substitution of less GHG-intensive materials for more intensive ones in various applications. The possible impact will vary with the GHG intensity, as well as, dependent on application, the substitution ratios of the concerned materials.

Several studies have during the last decades compared different building materials' energy and often also GHG gas balances, notably CO₂, either for themselves or when incorporated in various building constructions (see for instance Björklund & Tillman [8], Buchanan & Levine [9]). Upton et al. [10] review several earlier studies and in a summary they conclude for buildings relying on different construction materials: "For systems with comparable heating and cooling requirements wood-based building systems generally contain lower embodied energy and CO₂ emissions than steel-, concrete-, and brick-based systems." Gustavsson and collaborators have thoroughly compared the carbon balance implications for using wooden or concrete building frames in a life cycle perspective, including alternative

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options for land use and end-of-life management of the materials. They suggest that the substitution of wood for concrete in the structural frame of a specific four-storey residential building in Sweden, their study object, would give a significant net decrease in CO₂ emissions [11,12].

Put into the context of a climate policy, a higher price on CO₂ emissions also makes the wood framed building more competitive [13,14]. In the MATTER project (MATERials Technologies for greenhouse gas Emission Reduction), Kram et al. [15] modelled, in an integrated energy-materials-land use model for Europe, the effect of different CO₂ tax scenarios on the cost-effective substitution between materials in different sectors including buildings. They found an increase in the use of wood, while the demand for cement decreased when introducing a CO₂ tax. The project also pointed to that materials substitution could contribute with a considerable share of the total emission reductions and lower the costs compared to scenarios without materials substitution.

Summing up the results from previous studies in this field we find it fairly well-established that, given the current energy system, increasing the share of buildings with wood frames would reduce overall GHG emissions, and a few studies also point to that this could be a cost-effective strategy.

However, the cost-efficiency of different building frames over time from an energy system perspective in long-term scenarios meeting stringent climate targets has not been addressed. A long-term transition towards stringent CO₂ targets will, for certain, imply over the years a radically changing energy system and land use environment. Facilitating large decreases in the CO₂ intensity of the energy supply system will favour new technologies such as carbon capture and storage (CCS) and renewable energy, including bioenergy with possible implications for land scarcity. Such changes to the surrounding system may also change the effects of materials substitutions in the building sector. In this paper we use an energy systems model to generate two different climate stabilization scenarios: one scenario with carbon capture and storage (CCS) available, and one scenario that relies on rather expensive renewable energy sources. These scenarios have different effects on both the life cycle costs and carbon balances for different materials. We use these scenarios to analyse the following questions:

- Is it always the case that wood framed buildings have a better carbon balance than concrete framed buildings?
- Is substitution of wood frames for concrete frames a cost-effective climate mitigation strategy and how is the cost-effectiveness affected by technological development in the energy system?

The paper is structured as follows, in section 2 the methodology is described, in section 3 the results are presented, section 4 contains a discussion of the results and in section 5 conclusions are offered.

2. Method

In this paper we evaluate net present costs and carbon balances over the life-span of specific building frames with different materials composition. Both these factors are partly determined by the future energy system. The carbon dioxide emissions of buildings built in the future are dependent on the carbon intensity of the future energy system. Further the life cycle cost of a building is dependent on future energy prices and carbon dioxide prices. For this reason we use energy scenarios to evaluate buildings built today, as well as buildings built in the future.

2.1. Building frames

We use materials quantities from two functionally equivalent versions of a four-storey 1190 m² building (Wälludden, Sweden): a wood framed and concrete-framed version, see Table 1. This is the same building used in several previous studies on materials choices in building frames, e.g. Adalberth [16], Gustavsson et al. [17] and Dodoo et al. [12]. A more detailed description of the building is found in Persson [18]. This is a practical choice of building since it makes our study comparable to previous studies.

For each additional ton of wood in the wood frame (timber, plywood and particleboard), 40 tons of concrete is avoided.¹ This material substitution factor between wood and concrete for this building seems to be relatively high. Other studies have come up with substitution factors closer to 1:10 between wood and concrete. Therefore we also test (Section 3.3) the importance of this substitution factor by applying the same methodology to two other building frames, the Finnish 1175 m² Viikki building [17], and a 4032 m² hostel building from New Zealand [19,20].

2.2. System boundaries

Energy fluxes, CO₂ balances and costs included in the analysis are presented in Table 2. We make the analysis for two different construction years: 2010 and 2050. The buildings are assumed to have a use phase of 100 years and will hence be demolished in 2110 and 2150, respectively. We also construct scenarios where different energy technologies are available in the future, see section 2.3.

Energy use and costs for erecting the buildings and for operating and maintaining the buildings during the use phase (heating, lighting etc) are not included in the analysis. Following assumptions in previous studies on materials strategies in the building sector [16,17], these are assumed to be independent of the frame material. Calculations of the Wälludden building in Persson [18] also show identical energy use for the erection of the wood and concrete versions. However, for energy use during the use phase of the building even a small difference could indeed be important since the use phase is big in relation to the production phase. This assumption is further discussed in Section 4.

In the end-of-life phase some building materials may be recovered, which may translate to carbon dioxide flows as well as incomes for the project. Estimates of feasible recycling rates for building materials differ considerably in the literature. Dodoo et al. [12] assume 90% recycling rate for both wood and steel. Comparing to other studies [21–23] this seems to be fairly optimistic and in study we assume recycling rates of 80%. The sensitivity of the results to these rates is very low since the recycling occurs 100 years into the future when the CO₂ emissions of the surrounding energy systems are assumed to be low (see Section 2.3).

2.3. Energy scenarios

In order to construct consistent scenarios for prices of electricity, fossil fuels, biomass and carbon dioxide emissions we use a global energy systems model, GET, which minimizes the discounted energy system cost for the period 2000–2150 (see Azar et al. [24,25]; Hedenus et al. [26]). GET 7.0 consists of five demand sectors: transportation, electricity generation, industrial process heat, residential heat and feedstocks. In the model a large number of energy sources are represented: fossil fuels, nuclear energy, renewables etc, as well as a large number of energy conversion technologies.

¹ 2.5 tons of plasterboard is also added. The so-called wood building actually contains both more concrete and more plaster than wood in the building frame.

Table 1

Composition of the studied building frames from Wälludden Sweden. Given is also the composition for two other frames used in the sensitivity analyses.

Building Frame option		Main building frame		Frames used in sensitivity analysis			
		SWE Wälludden		FIN Viikki		NZ hostel	
		Wood	Concrete	Wood	Concrete	Wood	Concrete
Concrete	kg/m ²	187	1136	162	1714	284	692
Mortar	kg/m ²	20	19	0	0	0	0
Plasterboard	kg/m ²	75	21	118	19	34	34
Timber	kg/m ²	50	28	88	20	53	19
Particleboard	kg/m ²	15	14	23	8	11	0
Plywood	kg/m ²	18	17	13	0	2	0
Insulation	kg/m ²	18	8	20	8	0	0
Steel	kg/m ²	13	21	16	14	9	25
Concrete/wood substitution factor		40		16		8.8	

The advantage of using an energy systems model to generate prices is that the price relations between different energy carriers are consistent also in the long run. Since different energy sources are more or less good substitutes, the prices are interdependent. For instance natural gas and oil prices are related as both are used for process heat. The same applies for natural gas and coal as both are used for electricity generations. Such relations tend to change as the energy system is transformed over time. The energy system modelling approach also generates consistent bioenergy prices that include the increased costs due to land scarcity. Further, in GET 7.0 distribution and handlings costs of different energy carriers are included. However, the model has perfect foresight and the global energy market is assumed to work ideally. This results in that prices typically are lower than they would be in the real world. This is especially true for the oil price in the short run as OPEC market power is not represented. Thus, the model does not generate scenarios of the actual future, but it generates pictures of rather consistent possible futures. The scenarios could be seen as representing either a world that strives towards a 400 ppm stabilization target, or Europe that takes the lead in carbon mitigation in a world that strives for a less stringent carbon target.

Three energy supply system scenarios are generated: one baseline scenario (*Baseline*) without climate constraint and two scenarios where the atmospheric carbon dioxide concentration is stabilized at 400 ppm [26]. The two climate-constrained scenarios are named *Renewable* and *CCS* and differ with respect to the available energy technologies. In the *Renewable* scenario, shown in Fig. 1, carbon capture and storage technologies are not available and nuclear expansion is not allowed, which means that the

solution is dominated by renewable energy sources like solar and wind power. In the *CCS* scenario, shown in Fig. 2, carbon capture and storage is available from year 2020 [27] for large heat and power production plants as well as for industrial processes using both fossil fuels and bioenergy. This scenario also includes *CCS* for end-of-life management of wood materials (resulting in negative emissions). *CCS* is a technology under development, but plans are made to apply it to several sorts of industrial processes, for a good overview of the present state, see Haszeldine [27]. The different energy supply options affect the energy system in several ways. First, the carbon prices required to meet a 400 ppm scenario is higher in the *Renewable* scenario than in the *CCS* scenario, as *CCS* is assumed to be a cheaper technology than concentrated solar power at a large scale. Second, because *CCS* may be applied to industrial processes as cement and steel manufacturing, the carbon emission per unit of industrial heat produced is lower in the *CCS* scenario compared to the *Renewable* scenario. And third, the energy prices are lower in the *CCS* scenarios compared to the *Renewable* scenario.

As we consider houses built both in 2010 and 2050 this gives us, besides *Baseline*, four cases to be analysed, here called *Renewable 2010*, *Renewable 2050*, *CCS 2010* and *CCS 2050*, respectively. Wood or concrete framed building also affect the energy use in the world, as well as available bioenergy resources, even though the contribution is marginal compared to the overall system. In our system approach the buildings are added on top of the different scenarios, which means that the energy demand is slightly higher in the case of concrete framed buildings, and the use of bioenergy is marginally higher in the case of wood framed buildings.

Table 2Applied system boundaries. Energy use, CO₂ emissions, costs and land use changes included in the analysis.

Phase	Material	Energy flows included	CO ₂ flows included in carbon balance	Costs included	Assumed land/biomass use
Production	Wood	• electricity and heat for materials prod.	• energy related emissions	• materials prod. costs (incl. energy and CO ₂)	• felling of mature forest
	Concrete	• electricity and heat for materials prod.	• energy related emissions	• materials prod. costs (incl. energy and CO ₂)	• alternative land use: mature forest left with slow growth ^a
	Steel	• electricity and heat for materials prod.	• calcination emissions	• materials prod. costs (incl. energy and CO ₂)	
Use	Wood		• energy related emissions	• carbon uptake income	• growing of new forest
	Concrete		• iron reduction emissions	• carbon uptake income	
End-of-life	Steel		• uptake in new forest		
	Wood	• replacing other heat prod.	• uptake by carbonation		
	Concrete		• slow uptake in mature forest		
	Steel		• emissions from use in heat prod.	• heat replacement income	
	Wood		• avoided emissions for heat production	• CO ₂ emission cost	
	Concrete		• uptake by carbonation	• carbon uptake income	
	Steel		• material replacement	• materials replacement income	

^a For concrete frames less land is used, in this case we consider alternative land use, see section 2.4.

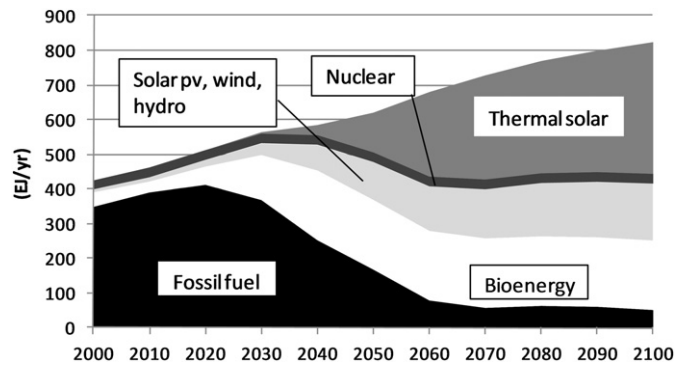


Fig. 1. Renewable scenario: GET run with atmospheric stabilization of CO₂ at 400 ppm without carbon capture and storage technology and no nuclear expansion.

2.4. Carbon balances

To calculate the net carbon emissions in 2010 we use the energy mix prevailing in European industry today. In 2050 we use the average carbon intensity in the industry sector for each scenario to calculate the carbon dioxide emissions, given the present energy requirements for the different material. However, as biomass residues are produced in the wood products industry, we assume that all heat for wood production in 2050 is produced from biomass in the 2050 scenarios. For that reason, the emissions intensity in the wood industry is lower than the emissions intensity in the steel and cement industries.

In the CCS 2050 scenario, apart from energy related emissions, we further assume that 80 percent of the carbon emission from the cement calcination process is captured [6]. Moreover, concrete structures are assumed to take up CO₂ through the carbonation process. The profile for the rate of carbonation over time is taken from Dadoo et al. [12]. Carbonation occurs at slow rate during the use phase of the buildings but accelerates after demolition of the buildings. In total the carbonation results in an uptake corresponding to 45 percent of the emissions through calcination.

Regarding land use, we assume that a new forest is planted on the forest land area harvested to provide the timber. Initially, the new forest grows at a rate of 1.7 ton C/ha/yr, but the growth rate slows down over time. This carbon sequestration is allocated as negative emissions (and thus in our cost estimate as an income when there is a price of carbon) for the timber used. The analysed concrete frame also contains a small amount of wood materials, but less than the wood frame. Hence, in the concrete case less land is utilized for materials use in the building sector. This unutilized land

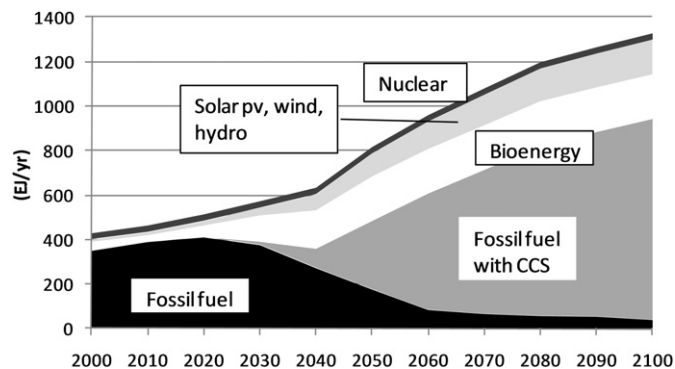


Fig. 2. CCS scenario: GET run with atmospheric stabilization of CO₂ at 400 ppm with carbon capture and storage technology but no nuclear expansion. The supply scenarios are all equal in the start year 2010.

could potentially be used for other purposes such as paper pulp or bioenergy. This is particularly likely in a world with stringent carbon policies and land scarcity [28]. However, we simply assume that this forest land is left as it is, with a slow growth rate of 0.2 ton C/ha/yr. These small amounts of sequestered carbon are allocated as negative emissions and hence give an income for the concrete option. In Section 3.3 we elaborate on the implications of another set of assumptions for land use.

2.5. Cost calculations

In order to estimate how the prices of different materials depend on changing energy prices, we begin by estimating a base cost for each material which is independent on energy and CO₂ prices. The prices of the different materials could then be written as a function of energy prices and CO₂ prices. We also isolate the price of material feedstocks for wood materials, assuming that this price follows the bioenergy price. Hence, if climate policy makes bioenergy a scarce resource, this scarcity rent is added also to the cost of wood materials. Eq. (1) describes how material prices are calculated.

$$P_{\text{tot}} = A + b \cdot p_{\text{bio_feedstock}} + c \cdot p_{\text{heat}} + d \cdot p_{\text{electricity}} + e \cdot p_{\text{CO}_2} \quad (1)$$

P_{tot} (€/t): Total price of a material at the year of construction

A (€/t): Base material cost: cost independent of energy and CO₂ prices

$p_{\text{bio_feedstock}}$ (€/t): Price of feedstocks for wood materials: dependent on bioenergy prices and hence CO₂ prices

p_{heat} (€/GJ): Heat price including CO₂ cost

$p_{\text{electricity}}$ (€/GJ): Electricity price including CO₂ cost

p_{CO_2} (€/t_{CO2}): emission price

The parameters A , b , c , d and e are presented with data in Table 3.

The price data for heat, electricity, bio_feedstock and carbon dioxide from the different scenarios is then used to calculate future cement, steel and wood materials prices as well as the value of future carbon dioxide emissions. From timber production, sawdust is produced as a rest flow, around 0.5 ton per every ton timber produced [29]. The sawdust constitutes a cheaper energy option for the forest products than imported energy sources. Even though sawdust is produced at low cost, the opportunity cost of bioenergy will increase. The difference in the energy price for sawmills compared to other industrial processes is the transportation cost of the residues to the market. We estimate this cost at 1 €/GJ [30,31]. In the demolition phase, 100 years after the building year, we assume that the wood materials in the building are incinerated for heat and thus generate an income according to the industrial heat prices in the different scenarios. Also the steel scrap generates income, at the price of 25 percent of primary steel [32].

Table 3

The parameters in the material costs equation based on Björklund & Tillman (1997), Sathre & Gustavsson (2007).

Material	A	b	c	d	e
	Base material cost	bio_feedstock	Heat	Electricity	Non-energy CO ₂
	€/t	t/t material	GJ/t	GJ/t	t CO ₂ /t
Concrete	41	0	0.76	0.08	0.07
Timber	620	1	3.54	0.49	0
Particleboard	330	1	7.94	1.51	0
Plywood	305	1	13.96	1.74	0
Plasterboard	259	0	7.94	0.58	0
Insulation material	1354	0	9.97	1.40	0
Steel	433	0	11.18	3.29	14.10

We also calculate the income due to carbon sequestration in forests and in the carbonation in the concrete. In the real world, it is unlikely that the house owner will be paid for this sequestration, but by including these payments we can study the social cost of the building. A discount rate of 5 percent is used in calculations of the net present cost.

3. Results

We begin in Section 3.1 by presenting the results of the model runs from the GET energy systems model and the resulting material prices. Section 3.2 contains the main results regarding carbon balances and costs for the wood and concrete framed versions of the main building of the analysis. Finally, a sensitivity analysis of these results is presented in Section 3.3.

3.1. Energy and materials prices

Fig. 3 presents results from the two energy supply scenarios *Renewable* and *CCS*. Both of these scenarios aim for stringent CO₂ targets of 400 ppm, but differ with respect to the available technologies: with or without CCS, see Section 2.1. The upper graphs show the generated outputs from the GET-model in terms of prices for CO₂ emissions, electricity, heat and bio_feedstocks (which increases due to land scarcity). The lower graphs show the resulting changes in materials prices (following Eq. (1)). As can be seen the effect on material prices are significantly lower in the *CCS* scenario compared to the *Renewable* scenario, and also the relative price differences between the materials are smaller in the *CCS* scenario.

The main difference between the two scenarios is that the availability of the CCS technology keeps down the price increase of energy in the *CCS* scenario which is also reflected in lower materials prices. The price peaks in 2060 in the *Renewable* scenario is the result of a very fast transformation in the modelled energy system at this point due to the prescribed carbon emission trajectory. This peak is of minor importance to the results, since we choose to analyse constructions built only in the years 2010 and 2050.

3.2. Carbon balances and costs for Wälludden concrete and wood frames

Fig. 4 shows different components of the carbon balance of the two Wälludden frames for the cases *Renewable 2010* and *CCS 2050*. These two cases represent two extremes as *Renewable 2010* uses the present energy system to construct the building, whereas the *CCS 2050* case uses a low carbon energy system and also uses CCS on the process emissions from steel and concrete production. Thus, the results of the other scenarios lie in between. The carbon emissions and uptake for sequestering in forest biomass, carbonation in concrete and end-of-life incineration of wood (for heat recovery) are equal in the two scenarios. The concrete frame causes more than twice as large emissions as the wood frame in the *Renewable 2010* case. For the *CCS 2050* case both frames result in net negative emissions which is possible due to an overall low emissions intensity in the industry sector, the application of CCS on cement production and BECCS on the wood utilization after demolishing of the house. CCS is

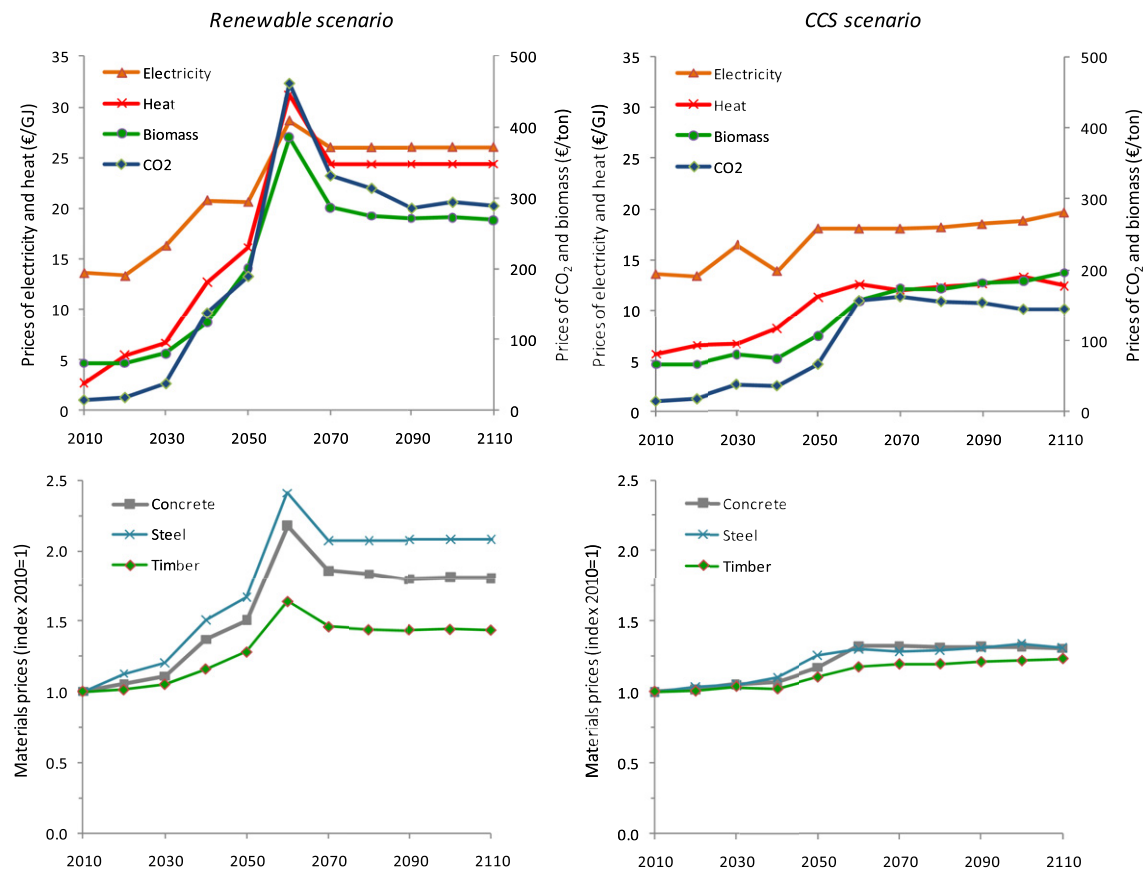


Fig. 3. The energy and materials and CO₂ prices in the two energy supply scenarios. The graphs to the left show results from the GET-model for the *Renewable* scenario and the diagrams to the right for the *CCS* scenario. The lower graphs present the resulting changes in materials prices in the two scenarios.

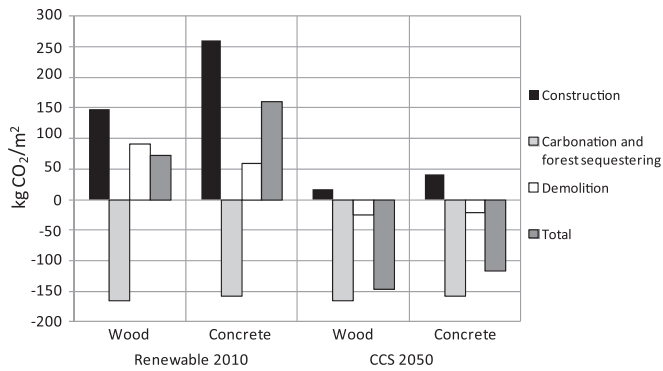


Fig. 4. The carbon balance of the wood and concrete frames in the *Renewable 2010* and *CCS 2050* cases.

assumed to capture 80 percent of the emissions from fuels and calcination of limestone, but during the life cycle of the building and after demolition concrete acts as a sink due to the carbonation process. The net result is negative emissions for concrete over the entire life cycle. As can be seen in Fig. 4 the wood frame has a better carbon balance in both scenarios. This result is, however, dependent on whether CCS is used for the combustion of wood materials after demolishing. We explore this assumption in the sensitivity analysis (Section 3.3).

A comparison of the net present cost of the materials in the two Wälludden frames with construction years at 2010 and 2050 is presented in Fig. 5 for three different supply cases: *Baseline* (only 2010), *Renewable*, and *CCS*. The net present costs have been calculated with a discount rate of 5 percent. The discount rate has a very little impact on the results in this analysis, though; discount rates in the range 2–8 percent give almost identical findings. The main reason for this is the dominance of the production phase in the total materials related costs.

The climate-constrained cases *Renewable 2010* and *CCS 2010* add very little to the overall cost of building frames. This is due to the dominance of the costs at the year of construction and that CO₂ and energy prices are still very low this early in the scenarios (Fig. 3). With a construction year at 2050, the CO₂ price is higher and this is also reflected in the costs of the two frames. Without CCS technologies, the cost of the concrete frame increases more than the wood frame, but with CCS the result is almost identical. The total cost difference between the two options is less than 1% in all cases except for *Renewable 2050* where it is 6%.

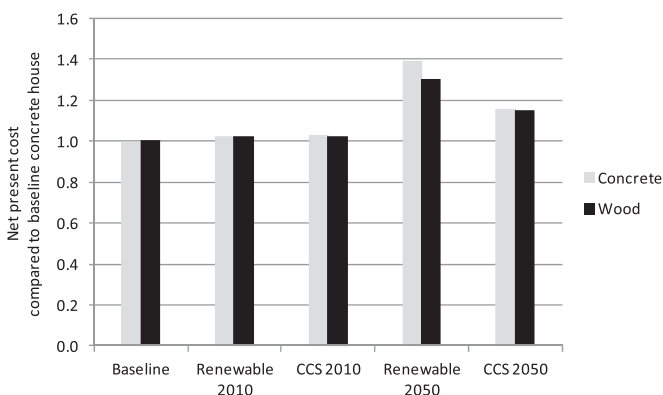


Fig. 5. Comparison of the net present cost of the materials in the Wälludden concrete and wood frames under different energy systems cases. The costs are indexed so that the net present cost of the concrete frame is 1 in the *Baseline* case, i.e. without climate policy.

3.3. Sensitivity analyses

In the previous section we presented results for carbon balances and costs, for a set of main assumptions, only varying between the energy systems scenarios (*CCS* and *Renewable*) and the year of construction (2010 and 2050). In this section we test the sensitivity of these results with respect to three other potentially important assumptions:

• Materials composition

In the previous section we presented results based on the material composition of concrete and wood framed versions of the Swedish Wälludden building. Here we apply the same analysis to two other buildings with functionally equivalent wood and concrete framed versions but with different concrete/wood substitution factors (Table 1).

• CCS for wood end-of-life building waste

In the main *CCS* scenario we showed that the wood frame causes lower carbon emissions than the concrete frame. This scenario assumes widespread applications of *CCS* throughout the energy system, including heat production from wood waste at the end-of-life of buildings. It is however unclear whether it is reasonable to assume that *CCS* would be used in small scale central heat plants which are the most likely receiver of wood waste. *CCS* is typically most cost-effective for large scale plants, and plants with *CCS* must also be located at a reasonable distance from a geological storage site. Here we test a more limited version of the *CCS* scenario, excluding *CCS* for building waste.

• Land use

The main assumption for land use in this study is that the forest which is felled for producing wood materials (for both frame options) is replaced with a new long-rotation forest for future production of wood materials. The lower wood demand for the concrete frame leaves an unused piece of land with a mature forest which is left as it is. This mature forest has a very slow growth rate and hence only minor uptake of CO₂. Here we instead assume that the forest on this land is also felled and used for some other purpose (e.g. paper pulp) and that the land is then used for short-rotation bioenergy production. The income generated by the old forest should not be included here. The bio-feedstock used for wood materials already implies a cost in the analysis and hence there should not be any “no-cost” biomass coupled to the option of the concrete frame. However, from a social planner perspective, it should be possible to generate income from the available clear cut land. We assume bioenergy plantations with a yield of 150 GJ/ha/yr (around twice as much as for regular long-rotation forests) and a production cost of 4 euro/GJ [33]. As climate policy makes biomass a scarce resource, the price rises above the production cost (see scenarios in Fig. 3) and starts to generate income. This can be seen as an alternative cost of using the land for timber.

3.3.1. Carbon balances

Fig. 6 summarizes the sensitivity analysis of these three assumptions for the carbon balance over the life cycle. The black bars show the *Renewable 2010* scenario and the grey the *CCS 2050* scenario. Positive values indicate higher emissions for the concrete frame. The main assumptions used in Section 3.2 correspond to the bars in the centre of the upper left diagram in Fig. 6. As seen if compared to Fig. 4, the total level of emissions is negative in the *CCS 2050* scenario, but positive in the *Renewable 2010* scenario.

End-of-life management of wood materials

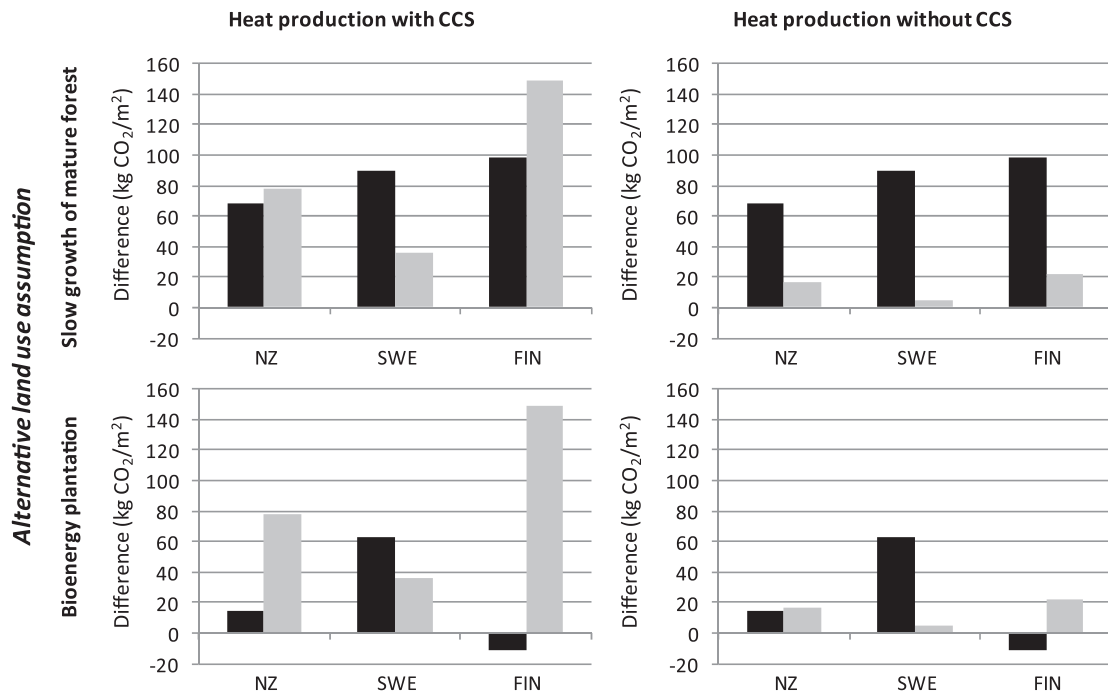


Fig. 6. The difference in CO₂ emissions per m² between a concrete framed building and a wood framed building. (Positive values thus indicate higher emissions for the concrete frame.) The black bars show the difference for the *Renewable 2010* scenario, whereas the grey bars are for the *CCS 2050* scenario.

Fig. 6 shows that the wood frames still have a better carbon balance in most cases, but the advantage diminishes in response to these changed assumptions. The results from the *Renewable 2010* scenario (black bars) are particularly sensitive to the land use assumption. The reason is that the energy system between 2010 and 2060 contains significant amounts of fossil fuels (see Fig. 1), which we assume are substituted by biomass grown on the spare land. After 2060 the energy system is almost carbon neutral and for that reason the carbon benefit diminishes for bioenergy plantations. This mainly affects the *CCS 2050* scenario. The land assumption further affects different building frames to various extents. Comparing the upper and the lower black bars (*Renewable 2010* scenario) shows that bioenergy plantations affect the Finnish frames to a much larger extent than the Swedish frames. The reason is that the Finnish wood frame contains more than 4 times as much wood materials as the concrete frame, whereas in the Swedish case the wood frame only contains around 50% more wood (Table 1).

The assumption on end-of-life use of wood matters only in the *CCS* scenario, since bioenergy with *CCS* is not available in the *Renewable* scenarios in any case. The grey bars in the lower right corner show that without *CCS* on wood waste, the carbon balances of wood and concrete frames become very similar.

3.3.2. Net present costs

In Section 3.2 we showed that the cost of the concrete and wood frame versions of the Wälludden building were almost identical in all scenarios except the *Renewable 2050* scenario (high carbon prices and no *CCS*), where the wood option had a lower cost (Fig. 5). Changing the materials composition to those in the Finnish Viikki and the New Zealand hostel building frames, which have much lower substitution ratios between concrete and wood (Table 1), gives a significant cost advantage to the concrete versions of the buildings.

Changing the land use assumption also gives an obvious advantage to the concrete option. Adding the income from

increased bioenergy production in the concrete option (or deducting it as an alternative cost for the wood option) means that the concrete frame gains in relative competitiveness in relation to the base case. With these assumptions, concrete frames become competitive also for high CO₂ prices and without *CCS* technology (*Renewable 2050*). Thus, despite the better carbon balance of the wood frames, concrete frames appears to be more attractive from a cost-efficiency perspective. However, as discussed below, the materials costs of a house, which we consider here, are only a minor part of the total cost of the building.

The *CCS* assumptions at the buildings' end-of-life have a very limited impact on net present costs since these costs occur in the future and are discounted over a 100 year period.

4. Discussion

It is important to note that both costs and emissions coupled to the building frame materials are small in relation to the building as a whole during its life cycle. In Fig. 7 we make a comparison of the involved costs. The total building cost is an average for Swedish residential buildings and includes all involved costs except for land [34]. The cost for frame materials accounts for only about 5 percent of the total construction cost. However, this is only materials costs for the frame, the total materials cost typically account for as much as 20 to 25 percent of the construction cost [35].

Even in the most extreme case of the *Renewable 2050* case, climate policy adds only around 40 percent to the costs of the building frames which is less than 2 percent of the total construction cost. Moreover, since both frames become more expensive with a climate policy, the cost difference is even smaller. In no scenario do the difference between the frames account for more than 0.2 percent of the total construction cost.

The energy use for heating during the use phase of the building is around 0.34 GJ/m²/yr or 34 GJ/m² during a life cycle of 100 years

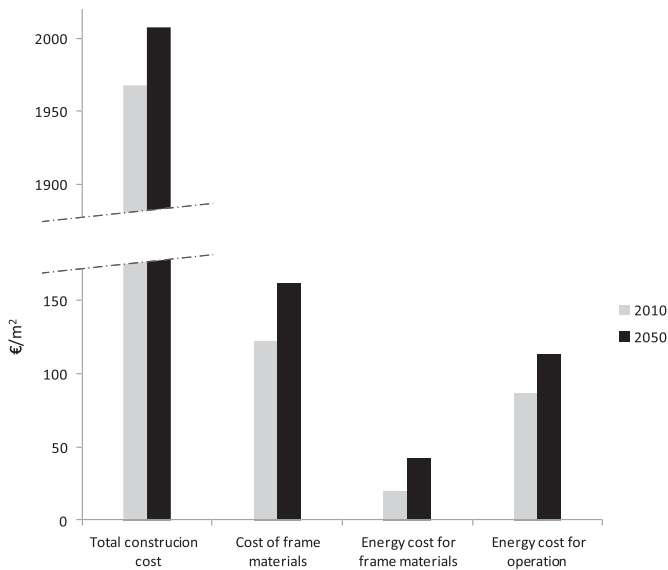


Fig. 7. Comparison of different costs in a buildings life cycle. The energy cost of frame materials is a sub-set of the cost of frame materials which in turn is a sub-set of the total construction cost. All costs are taken from the Renewable scenario and represent the average of the wood and concrete frames. The energy cost for heating during 100 years of operation is calculated with a discount rate of 5%.

[16]. The heat and electricity demand for producing the materials of the building frame is 1.68 GJ/m² for the wood frame and 2.26 GJ/m² for the concrete frame.² The concrete and wood framed versions of the building have been reported to be functionally identical. For example, Gustavsson and Sathre [36] refer to calculations in Adalberth [16], which show that the difference in the use phase is less than 1 percent (Adalberth's modelling showed that the energy use in the concrete framed version is 0.7 percent lower than in the wood framed version). This is indeed a small difference. However, when compared to the energy use in the production of the frame materials, a difference of 1 percent in the use phase is not insignificant. This corresponds to 0.34 GJ/m² or almost 60% of the difference between the production of the wood and the concrete frames (0.58 GJ/m²). In terms of costs, the energy use in the use phase accounts for a smaller share due to the discounting of future cost. At a discount rate of 5 percent, the energy costs during operation of the building is similar to the materials cost of the building frame, see Fig. 7.

To sum up, if the choice of building frame materials in any way affect costs and emissions outside of the system boundaries of this analysis (see Table 2), such as other costs in the production process or energy use, maintenance costs or renovation costs later in the life cycle this may significantly affect both cost-effectiveness and total CO₂ emissions.

5. Conclusions

This study has focused on some aspects of the building sector adaptation to long-term stringent CO₂ targets. We have investigated the possible changes in energy, CO₂ and cost characteristics of major building frame materials due to adaptation to stringent climate goals by using a global energy systems model.

² These figures are for building frame materials only. The total energy use for building materials is typically around 3 GJ/m². With the very wide system boundaries of an input–output analysis including all transport and other upstream activities (services, production of machinery etc), the total energy use for production is as high as 6 GJ/m² [4].

Our analysis confirms the results from previous studies that for current conditions wood framed buildings will emit less CO₂ during their life cycle than concrete buildings. Built on these earlier results arguments have been put forward that using wood frames in buildings should be stimulated. Still, in an elaborative scenario where CCS technologies are made available in the energy system, the better carbon balance for wood frames is dependent on that CCS would also be used for the incineration of the relatively small and scattered streams of wood waste at the end-of-life of the building. Moreover, the carbon balances are sensitive to assumptions on the alternative use of the forest land in the concrete case, for which the land requirements for materials production are smaller than in the wood case.

Our analysis also shows that the difference in cost of materials (including a price on CO₂ emissions) between a wood frame building and a concrete frame building is very small and ambiguous in most scenarios, and for one of the buildings analysed actually the materials cost for the concrete frame was significantly lower. The material cost is also just a part of much larger total construction costs of the buildings, in which they are embedded and possibly will have an influence on. Thus, even if wood frames have a better CO₂ balance in most scenarios they are not necessarily a cost-effective mitigation option even in the long run. For that reason, a more detailed cost analysis should be made before policymakers start to design policies to promote wood frames. Biomass is a limited resource which may be more cost-effectively used for climate mitigation in other sectors of the economy, for example in the energy sector.

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